

2017 年概率论与数理统计试题答案

一、填空题

$$\begin{aligned} &1、0.75 \quad 2、\frac{1}{2} \quad 3、\begin{cases} f_X\left(\ln\frac{y}{3}\right)\frac{1}{y}, & 0 < y < +\infty \\ 0, & \text{其他} \end{cases} \quad 4、\frac{5}{16} \quad 5、1 \\ &6、(1-e^{-1})^2 \quad 7、\frac{2}{45} \quad 8、\frac{1}{2(n-1)} \quad 9、t(9) \quad 10、(4.134, 6.266) \end{aligned}$$

二、解：设事件 A 表示产品来自工厂 A，事件 B 表示产品来自工厂 B，事件 C 表示抽取到的产品是次品，则

$$P(C|A)=1\%, \quad P(C|B)=2\%, \quad P(A)=60\%, \quad P(B)=40\%$$

$$\text{从而 } P(A|C) = \frac{P(A)P(C|A)}{P(A)P(C|A)+P(B)P(C|B)} = \frac{60\% \cdot 1\%}{60\% \cdot 1\% + 40\% \cdot 2\%} = \frac{3}{7}$$

三、解：(1) 由概率密度函数的性质

$$1 = \int_{-\infty}^{+\infty} f(x)dx = \int_0^{\frac{1}{2}} (cx^2 + x)dx = \frac{c}{24} + \frac{1}{8}$$

解得 $c = 21$

$$(2) \text{ 由 } f(x) = \begin{cases} 21x^2 + x, & 0 \leq x \leq \frac{1}{2}, \text{ 则} \\ 0, & \text{其他} \end{cases}$$

$$F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} 0 & x < 0 \\ \int_0^x (21t^2 + t)dt = 7x^3 + \frac{x^2}{2} & 0 \leq x \leq \frac{1}{2} \\ 1 & x > \frac{1}{2} \end{cases}$$

$$(3) \quad P\left(X \leq \frac{1}{3}\right) = F\left(\frac{1}{3}\right) = \frac{17}{54}$$

四、解：由卷积公式

$$f_{X+Y}(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx \quad 3 \text{ 分}$$

$$\text{易知仅当 } \begin{cases} 0 \leq x \leq 1 \\ z-x > 0 \end{cases} \quad \text{即} \quad \begin{cases} 0 \leq x \leq 1 \\ x < z \end{cases}$$

时被积函数不为零

2 分

$$f_{X+Y}(z) = \begin{cases} 0, & z < 0 \\ \int_0^z e^{-(z-x)} dx & 0 \leq z < 1 \\ \int_0^1 e^{-(z-x)} dx & z \geq 1 \end{cases}$$

$$\text{即 } f_{X+Y}(z) = \begin{cases} 0, & z < 0 \\ 1 - e^{-z} & 0 \leq z < 1 \\ e^{-z}(e-1) & z \geq 1 \end{cases}$$

$$\text{五、解: (1) } f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^1 6xy^2 dy = 2x & 0 < x < 1 \\ 0 & \text{其他} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^1 6xy^2 dx = 3y^2 & 0 < y < 1 \\ 0 & \text{其他} \end{cases}$$

显然, $f(x, y) = f_X(x)f_Y(y)$, 所以 X, Y 相互独立

$$(2) \text{ 当 } 0 < y < 1 \text{ 时, } f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} 2x, & 0 < x < 1 \\ 0, & x \text{ 取其他值} \end{cases}$$

$$\text{六、解: (1) } E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyf(x, y) dx dy = \int_0^1 dx \int_0^x 3x^2 y dy = \int_0^1 \frac{3}{2} x^4 dx = \frac{3}{10}$$

$$E(X) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} xf(x, y) dy = \int_0^1 dx \int_0^x 3x^2 dy = 3 \int_0^1 x^3 dx = \frac{3}{4}$$

$$E(Y) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} yf(x, y) dy = \int_0^1 dx \int_0^x 3xy dy = \frac{3}{2} \int_0^1 x^3 dx = \frac{3}{8}$$

$$\text{则 } \text{cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{3}{160}$$

$$(2) E(X^2) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} x^2 f(x, y) dy = \int_0^1 dx \int_0^x 3x^3 dy = 3 \int_0^1 x^4 dx = \frac{3}{5}$$

$$E(Y^2) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} y^2 f(x, y) dy = \int_0^1 dx \int_0^x 3xy^2 dy = \int_0^1 x^4 dx = \frac{1}{5}$$

$$D(X) = E(X^2) - E^2(X) = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{80}$$

$$D(Y) = E(Y^2) - E^2(Y) = \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{19}{320}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = \sqrt{\frac{3}{19}}$$

七、解: 设 X 表示正常工作的部件个数, 则 $X \sim B(100, 0.9)$, 由棣莫弗-拉普拉斯

定理,

$\frac{X-100 \times 0.9}{\sqrt{100 \times 0.9 \times 0.1}}$ 近似服从 $N(0,1)$ 分布,

则

$$P(X \geq 84) = 1 - P(X < 84) = 1 - P\left(\frac{X-90}{3} < \frac{84-90}{3}\right) \approx 1 - \Phi(-2) = \Phi(2) = 0.9772$$

八、解: (1) 由于

$$E(X) = \int_0^{+\infty} x \frac{\theta^2}{x^3} e^{-\frac{\theta}{x}} dx = \int_0^{+\infty} \frac{\theta^2}{x^2} e^{-\frac{\theta}{x}} dx = \theta \int_0^{+\infty} e^{-\frac{\theta}{x}} d\left(-\frac{\theta}{x}\right) = \theta,$$

令 $\theta = \bar{X}$, 解得 θ 的矩估计量为 $\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

(1) 似然函数为

$$L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \begin{cases} \prod_{i=1}^n \frac{\theta^2}{x_i^3} e^{-\frac{\theta}{x_i}}, & x_i > 0 (i=1, 2, \dots, n) \\ 0, & \text{其他.} \end{cases}$$

当 $x_i > 0 (i=1, 2, \dots, n)$ 时, $L(\theta) = \prod_{i=1}^n \frac{\theta^2}{x_i^3} e^{-\frac{\theta}{x_i}}$, 两边取对数

$$\ln L(\theta) = \sum_{i=1}^n \left[2 \ln \theta - \ln x_i^3 - \frac{\theta}{x_i} \right]$$

$$\text{令 } \frac{d \ln L(\theta)}{d \theta} = \sum_{i=1}^n \left[\frac{2}{\theta} - \frac{1}{x_i} \right] = \frac{2n}{\theta} - \sum_{i=1}^n \frac{1}{x_i} = 0, \quad \theta \text{ 的最大似然估计量为 } \hat{\theta} = \frac{2n}{\sum_{i=1}^n \frac{1}{X_i}}, \text{ 解得}$$